

Munim Thahmid

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Research Interests

Reliable and secure software systems; agentic SRE; production failure analysis; cloud/distributed systems; AI-agent safety and security; software reliability and testing.

Education

Bangladesh University of Engineering and Technology

B.Sc. in Computer Science and Engineering; CGPA: 3.46/4.00

Dhaka, Bangladesh

Jan. 2022 – May 2026

Research Experience

Summer Research Intern, University of Illinois Urbana-Champaign

Remote

Advisor: Prof. Tianyin Xu — Safe and Reliable Agentic SRE

2026 – Present

- Studying how agentic SRE systems diagnose, mitigate, and validate real-world production failures in cloud and Kubernetes-style environments.
- Contributed a reproducible SREGym failure scenario for node conntrack exhaustion, including runtime setup, oracle logic, and analysis of why the failure blocks normal diagnostic access paths.
- Analyzing production failures through the lens of agent capabilities: context management, uncertainty handling, safe action execution, rollback, validation, and cost-efficient diagnosis.

Undergraduate Thesis Research, BUET

Dhaka, Bangladesh

Supervisor: Dr. Sadia Sharmin — Banglish Robustness in Bangla LLMs

2025 – 2026

- Built a paired Bangla/Banglish/English benchmark from BEnQA and BanglaMATH, preserving item IDs and gold answers to isolate script choice as the main robustness variable.
- Evaluated compact open-weight and hosted API models, showing that Latin-script Banglish can remain less reliable than native Bengali script even when the underlying task and answer are unchanged.
- Added review sensitivity, tokenization analysis, paired recoverability analysis, a 974-row human-reviewed BEnQA extension, natural code-mixed validation, and negative mitigation experiments.

Publications and Manuscripts

Layer-wise Probing of Whisper’s Encoder Representations for Bengali Phone-like Units

Munim Thahmid, Sadia Sharmin. *Accepted to INTERSPEECH 2026.*

- Probed Bengali phone-like unit separability across Whisper-small, Whisper-medium, and Whisper-large-v3 under speaker-disjoint evaluation.
- Found mid-to-late layer peaks and reduced final-layer degradation at larger scale; validated trends using XLSR baseline, ABX discriminability, confidence filtering, duration stratification, and probe ablations.

Script Matters: Measuring Latin-Script Banglish Robustness in Bangla LLMs

Munim Thahmid, Sadia Sharmin. *B.Sc. Thesis / Manuscript, 2026.*

- Introduced a controlled paired evaluation protocol showing that script choice can change LLM reliability for semantically equivalent Bangla tasks.
- Showed that the Banglish gap is not explained by simple cleanup or token length alone, and that naive normalization/generation-based mitigation remains brittle.

Technical Skills

Programming: Python, C/C++, JavaScript/TypeScript, Bash, SQL

Systems: Linux, Docker, Kubernetes, Git, SREGym, Kind, cloud/debugging workflows

ML/NLP/Speech: PyTorch, HuggingFace Transformers, Whisper, wav2vec2/XLSR, scikit-learn, pandas, NumPy

Research: Benchmark construction, statistical evaluation, bootstrap confidence intervals, error analysis, reproducibility artifacts